

Navya Bulusu

+1 469-939-9096 | navyabulusu@gmail.com | <https://www.linkedin.com/in/navyabulusu/> | <https://navyabulusu.netlify.app/> | <https://github.com/navyaa97> | <https://medium.com/@navyabulusu>

PROFESSIONAL SUMMARY:

Insight-driven **Senior Data Analyst** with 5+ years of experience transforming complex data into intuitive, story-driven insights across **Vanguard** and **Applied Materials**. Known for designing data ecosystems that simplify decision-making, connect business and technology, and inspire executive action. Proficient in **Power BI, Tableau, Python, SQL, and cloud platforms (AWS, Azure)** with a strong focus on predictive analytics, automation, and user-centered dashboard design. Passionate about using data to elevate business strategy, enhance product experiences, and drive measurable impact at scale.

Key Achievements:

- Delivered \$4M+ in cumulative business impact through analytics automation, predictive forecasting, and executive storytelling.
- Designed Power BI dashboards that reduced reporting time by 30% and improved leadership visibility across operations and cost centers.
- Partnered with executives to translate analytics into strategic business actions—reducing logistics costs by 20% and improving data reliability by 35%.

TECHNICAL SKILLS:

- | | |
|------------------------------|--|
| • Programming: | GenAI, Python (NumPy, Pandas, Scikit, Matplotlib, Seaborn, TensorFlow), R, VBA, Impala, Java, JSON |
| • ETL and Reporting: | Tableau, Power BI, MS Excel (VBA, macros), Power Query, Snowflake, SAS, AWS, Hadoop, Airflow |
| • Databases and SDLC: | SQL, MySQL, MS SQL Server, MongoDB, Agile, SAP, Waterfall |

CORE COMPETENCIES:

Business Storytelling • Dashboard Design • Time Series Forecasting • Data Strategy & Governance • Executive Analytics • KPI Frameworks • Cross-Functional Collaboration • Supply Chain Analytics • Data-Driven Decision Making

CERTIFICATIONS:

Databricks Generative AI, Snowflake SnowPro CORE, Tableau Desktop Specialist, IBM Data Analysis with Python, CAPM – Dec 2025

BUSINESS EXPERIENCE

Vanguard, Dallas, TX | Data Analyst

August 2025 – Present

- Partnered with cross-functional teams to **translate operational challenges into analytical solutions**, developing frameworks that aligned data with business goals across the **Self-Directed Investor Experience (SDI)** and **Lead Generation** programs.
- Designed and maintained **Power BI/Tableau dashboards** integrating **Amazon Athena, AWS S3, and ServiceNow** data sources, applying **dataflows, DAX, and KPI design principles** to track **NCH, CSAT, and cost-center performance**—reducing manual reporting time by **30%** and improving accuracy across departments.
- Developed **predictive and time-series models in Python** to forecast call volume and escalation trends, enabling proactive staffing strategies that decreased customer wait times by **18%**.
- Conducted **NLP-based analysis on ServiceNow tickets** to categorize recurring support issues, improving triage efficiency by **40%** and providing leadership with insights into top customer pain points.
- Delivered **executive-ready storyboards** with drill-through narratives, scenario modeling, and “What-If” parameters—turning performance data into **strategic decision tools** that shaped quarterly planning sessions.

Applied Materials, Austin, TX | Data Scientist

March 2023 – July 2025

- Applied advanced **AI/ML** techniques to analyze complex datasets, developing robust models that optimized routing and delivery schedules, reducing **logistics costs by 20%** and **improving delivery times by 15%**.
- Designed and optimized **LLM and NLP frameworks** to streamline supply chain communications and automate data entry, enhancing decision-making processes and improving **operational efficiency by 25% (GenAI)**.
- Independently built **RAG** applications using open-source LLM models, enhancing data retrieval and analysis capabilities, leading to a **30%** increase in informed decision-making and operational efficiencies.
- Utilized **SQL** to efficiently coordinate **data extraction**, manipulation, and querying processes, ensuring **data integrity** and precision throughout analysis, which led to a **35%** increase in data quality and a **25%** boost in processing speed.
- Leveraged **Cloudera, Python, and Hadoop** ecosystems to streamline processing of large-scale datasets, slashing processing time by 50% and boosting **data throughput by 60%**.
- Engineered and implemented Python-based **machine learning** models including demand forecasting using **ARIMA**, route optimization employing genetic algorithms, and **cost prediction models** with **regression** techniques. Achieved a **25% reduction in delivery times** and a **20% increase in cost efficiency** in logistics operations.

Applied Materials, Austin, TX | Business Intelligence Analyst**Aug 2021 – March 2023**

- Worked as a Business Intelligence Analyst in the **Supply Chain** department, handling Inventory, Demand, and Purchasing data to improve inventory **accuracy** by **20%** and reduce **demand forecasting** errors by **15%**.
- Executing an efficient way to automate reporting using Power Query, DAX and Macros/VBA saving **12%** of time; Engaged in **ETL** process from **Hadoop** Data center and **SAP** to **SQL** Server to handle Big Data.
- Involved in developing **KPIs** and metrics to evaluate each supplier performance using Tableau and Power BI Dashboards; introduced new evaluation system using **predictive analytics**.
- Leveraged **Snowflake** to manage and analyze **large datasets in the cloud**, improving data accessibility and processing speed, which enhanced overall **reporting efficiency** by **15%**.
- Utilized **Azure Databricks** for **advanced data processing** and **machine learning** model development, resulting in improved **forecasting** accuracy and optimized supply chain operations.

Applied Materials, Santa Clara, CA | Data Analyst Intern**July 2020 – Sept 2020**

- **Saved** about **20%** productivity time for the **Product line Management team** by collecting data from multiple resources (**ETL**) using HDFS and achieving data integration by performing data mapping and providing recommendations to the sales executives to automate the process.
- Additionally, performed extensive **Exploratory Data Analysis (EDA)** using **Pandas**, identifying key trends and patterns, and provided actionable recommendations to sales executives to automate processes.

Amazon, Hyderabad, Telangana, India | Transaction Risk Investigator**Feb 2019 – April 2019**

- Played an instrumental role in providing a fraud free e-commerce platform to the customers by evaluating and analyzing 300+ daily transactions and identifying potentially fraudulent activities with **96%** accuracy.
- Improved data accuracy and reporting integrity by managing operational databases of 10 – 15 TB using **MySQL** to create reports and retrieve as well as alter the data in alignment with the business requirements.

Kinectra Pvt Lmt., Hyderabad, Telangana, India | Jr. Data Analyst**May 2018 – Dec 2018**

- Supported the identification of potential market and creation of business and go-to-market strategies by conducting extensive market research, interpreting data, extracting actionable insights, and providing ongoing reports to the management.
- Organized a large data set of over **30 TB** and predicted the behavior of the given data using machine learning algorithms, performing data analysis, understanding business requirements and strategies by performing **data wrangling**.

PROJECTS**Prediction of GPU runtimes** (Python – numpy, pandas, scikit, matplotlib)

- Experimented datasets with both supervised and unsupervised learning, implemented **linear** and **logistic regression** models on the dataset, and tested model's performance by applying **gradient descent**, cost function with batch updates and by experimenting on hyperparameters.
- Implemented decision trees and neural network packages by converting into classification problem; and clustering techniques like **K – means**; Expectation maximization and compared the performances of each algorithm using learning curves and confusion matrices.
- Improved model performance and accuracy by implementing feature reduction algorithms like **principal component analysis (PCA)** and independent component analysis (ICA) and choosing the best one.

Prediction of Housing Prices in Boston (R Programming)

- Accurately predicted the cost of houses in particular area based on various parameters from operated the dataset using supervised machine learning like regression algorithms.
- Pre-processed data using scaling and normalization to transform the data into the similar range and executed exploratory data analysis for prediction of accurate market price of the house reliant on those factors.

SQL- MongoDB – Spatial data of wildfires in US (MySQL and MongoDB)

- Built a physical database using the spatial data of US wildfires through performing **data modeling**, loading the data into the database, and carrying out SQL queries using MySQL workbench while implementing joins, indexing, and other basic queries.
- Developed Mongo Database and transferred the data from MySQL workbench to MongoDB and performing Mongo Queries.

EDUCATION**Campbellsville University**

Masters in Information Technology Management

*December 2025***The University of Texas at Dallas**

Masters in Business Analytics

*August 2021***Jawaharlal Nehru Technological University, Hyderabad, India**

B.Tech., Computer Science and Engineering

April 2018